

**Example.** (Stress tensor) Let the integrand be denoted

$$I = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \alpha R\phi^2$$

then,

$$\begin{aligned} T_{\mu\nu} &= -\frac{2}{\sqrt{-g}}\frac{1}{\delta g^{\mu\nu}}\delta(\sqrt{-g})I - \frac{2}{\sqrt{-g}}\frac{1}{\delta g^{\mu\nu}}\sqrt{-g}\delta I \\ &= g_{\mu\nu}I - \frac{2}{\delta g^{\mu\nu}}\left[-\frac{1}{2}\partial_\mu\phi\partial_\nu\phi\delta g^{\mu\nu} - \alpha\delta g^{\mu\nu}R_{\mu\nu}\phi^2 - \alpha g^{\mu\nu}\delta R_{\mu\nu}\phi^2\right] \\ &= -\frac{1}{2}\partial_\mu\phi\partial_\nu\phi + \alpha\phi^2\left[2R_{\mu\nu} - g_{\mu\nu}R + (g_{\mu\nu}g^{\rho\sigma}\nabla_\rho\nabla_\sigma - \nabla_\mu\nabla_\nu)\right] \\ &= -\frac{1}{2}\partial_\mu\partial_\nu\phi + 2\alpha\left[\phi^2G_{\mu\nu} + (g_{\mu\nu}g^{\rho\sigma}\nabla_\rho\nabla_\sigma - \nabla_\mu\nabla_\nu)\phi^2\right] \end{aligned}$$

In going from lines 2 to 3, we have used the variation of a matrix determinant and the Palatini's identity. Also, from lines 3 to 4, we have used integration by parts as we can assume that the expression is under the integral sign. On-shell, notice that the expression's trace is

$$\text{Tr}(T) = -\frac{1}{2}(\nabla\phi)^2 + 2\alpha\left[\phi^2\left(-\frac{D-2}{2}R\right) + (D-1)g^{\rho\sigma}\nabla_\rho\nabla_\sigma\phi^2\right]$$